Newton Adhikari

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PROFESSIONAL SUMMARY

Computer Science and Engineering student with experience in Artificial Intelligence, Machine Learning, Reinforcement Learning, and Full-Stack Development. Skilled in building AI-powered systems, scalable web applications, REST APIs, and database-driven solutions using Python, Go, React.js, PostgreSQL, and OpenAI APIs. Published researcher and patent holder with leadership experience in technical communities and innovation-driven projects.

EDUCATION

- **Guru Nanak Institute of Technology, Hyderabad, India** **(2023 – Present)**
  Bachelor of Technology in Computer Science and Engineering
  Affiliated with Jawaharlal Nehru Technological University Hyderabad (JNTUH)
- **Caspian Valley Higher Secondary School, Kathmandu, Nepal** **(2020 – 2022)**
  Science Stream (Physics, Chemistry, Mathematics, Biology)
  Affiliated with the National Examination Board(NEB), Nepal
- **Shree Kishori Janata Secondary School, Hariharpur Harinmari, Nepal** **(2012 – 2019)**
  Secondary Education (Class 10)
  Affiliated with the National Examination Board(NEB), Nepal

PROJECTS

- **AI-Powered Virtual Assistant** *Python, OpenAI API*
  Developed a voice-enabled AI assistant with speech recognition, task automation, web search, and application control.
- **A Hybrid Reinforcement Learning for Supply Chain Optimization** *Python, PPO, CTDE*
  Built a hybrid RL framework for inventory management, demand forecasting, and supply chain optimization.
- **Education Operating System (EduOS)** *React.js, Go, PostgreSQL*
  Developed a role-based education management platform with attendance tracking, academic modules, and REST APIs.
- **AI-Based GPS Tracking System** *OpenAI API, Maps API*
  Implemented an AI-powered GPS tracking solution with real-time monitoring, route analysis, and smart navigation insights.
- **Secure Email Authentication System** *React.js, Go, PostgreSQL*
  Built a secure authentication system featuring email verification, password recovery, JWT authentication, and role-based access control.

RESEARCH, PUBLICATIONS & PATENTS

- **Patent**:A Hybrid Reinforcement Learning for End-to-End Supply Chain Management.
- **Published Research Paper**: Performance and Security Analysis of Blockchain-Based Decentralized Cloud Storage Systems. DOI: https://doi.org/10.63949/crinfo.v2i1.004
- **Research Paper Publication**: Decision-Calibrated Probabilistic Learning for Cost-Sensitive and Risk-Aware Decision Making.

EXPERIENCE & LEADERSHIP

- **President, HackerRank Club (CSE Department)** **Guru Nanak Institute of Technology**
  - Led technical initiatives promoting coding, innovation, and problem-solving among students.
  - Organized **MetaStorm**, **CreateX**, and **HackFun Hackathon**, managing event planning, coordination, and participant engagement.
- **Member, Google Developer Club (GDC)** **Guru Nanak Institute of Technology**
  - Participated in technical workshops, developer events, and hands-on learning sessions.
  - Collaborated on technology-focused initiatives and contributed to the campus developer community.

SKILLS

- **Programming Languages**: Python, Java, C, JavaScript, Dart, HTML, CSS, Golang
- **Machine Learning & Deep Learning**: Classification, Regression, Clustering, Convolutional Neural Networks (CNN), Long Short-Term Memory Networks (LSTM)
- **Data Analysis & Visualization**: Pandas, NumPy, Matplotlib, Seaborn
- **Frameworks & Technologies**: Flutter, React.js, Node.js, Express.js, REST APIs, Firebase, Bootstrap
- **Databases**: MySQL, PostgreSQL
- **Developer Tools**: Git, GitHub, Visual Studio Code, Android Studio, Bruno, Figma
- **Core Computer Science**: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks, Software Engineering
- **Areas of Interest**: Artificial Intelligence & Machine Learning, Mobile App Development, Web Development, UI/UX Design, Cloud Computing
- **Languages**: English, Nepali, Hindi, Maithili, Bhojpuri

CERTIFICATIONS & ACHIEVEMENTS

- **Certificate of Achievement** – Organized and led **MetaStorm**, a technical innovation event under the HackerRank Club, GNIT.
- **Certificate of Achievement** – Organized **CreateX**, a technology and creativity-focused event promoting innovation and problem-solving.
- Participated in **HackArena National Hackathon**, demonstrating collaborative problem-solving and software development skills.
- Participated in the **PyQuest Technical Quiz Competition** conducted by the Data Pioneer Club.
- Successfully completed a **Virtual Internship Program** with **CodSoft**, gaining practical software development experience.
- Attended the **Emerging Trends and Cyber Security Challenges** workshop organized by the Department of Computer Science and Engineering, GNIT.
- Participated in the **Brainic Career Guidance Webinar** focused on professional development and career planning.